Automated Business Sales Reporting with n8n

Project Overview

This project automates sales reporting for a business using n8n, Google Sheets, Gemini AI, and Gmail.

The workflow retrieves sales transaction data, calculates key financial and operational metrics, compares current-period revenue with the previous period, and generates an AI-assisted email report when revenue drops by more than 20%.

Business Problem

Preparing sales reports manually can take time and make it harder for business owners to identify changes in performance quickly.

This automation helps reduce repetitive reporting work and alerts the business when revenue falls significantly.

Tools and Technologies

- **n8n:** Workflow automation and orchestration
- **Google Sheets:** Sales data source and report log
- **JavaScript:** Data processing and financial calculations
- **Google Gemini API:** AI-generated report and recommendations
- **Gmail:** Email delivery

Key Features

- Retrieves sales records from Google Sheets.
- Calculates revenue, order count, average order value, cost of goods sold, gross profit, gross margin, line items, and units sold.
- Compares revenue across consecutive seven-day periods.
- Checks whether revenue has declined by more than 20%.
- Generates an AI-assisted report with recommendations when the alert condition is met.
- Sends the report through Gmail.
- Records report metrics in a Google Sheets worksheet.

Workflow

1. Manual or scheduled trigger starts the workflow.

2. Business reporting parameters are prepared.
3. Sales records are retrieved from Google Sheets.
4. Sales data is processed and reporting metrics are calculated.
5. An IF node checks the revenue-change threshold.
6. When the threshold is exceeded, Gemini generates the report content.
7. Gmail sends the report to the configured recipient.
8. Google Sheets records the report metrics.

Testing and Validation

The workflow was executed manually using sample sales data.

The test produced the following results:

- Current-period revenue: KES 238,085.25
- Previous-period revenue: KES 352,969.25
- Revenue change: approximately -32.55%
- Revenue-drop condition: True
- Email report: Successfully sent
- Report log: Successfully updated

These results demonstrate that the tested revenue-alert path executed successfully. Additional testing is required for other scenarios, including periods without a revenue drop and scheduled execution.

Potential Business Value

This workflow can help small businesses monitor sales performance, identify significant revenue changes, and reduce repetitive reporting tasks.

It can be adapted to different businesses by adjusting the data schema, reporting metrics, alert thresholds, and email recipients.

Privacy and Security

This demonstration should use synthetic or anonymized sales data. API credentials, passwords, private customer information, and other sensitive data must not be published.

Future Improvements

- Support configurable reporting periods.
- Send routine reports even when revenue does not trigger an alert.
- Add product-level and category-level performance insights.
- Improve error handling and data validation.

- Test scheduled execution and duplicate-report prevention.
- Add a dashboard for sales trends and profitability.